Waipapa Taumata Rau
University of Auckland

What can ENGGEN403 do for me?

A guide to making the most of this course

July 26

Juliet Gerrard

Gnarly real world, real problems

BUILDS ON PREVIOUS ENGGEN COURSES

TEACHES YOU TO THINK ABOUT ACTUAL SYSTEMS PROBLEMS OF NATIONAL SCALE

DIRECT CONNECTION TO GOVERNMENT

BUILDS ESSENTIAL WORKPLACE SKILLS THAT COMPLEMENT YOUR TECHNICAL KNOWLEDGE

HELPS WITH THAT INTERVIEW ...
...TO GET A JOB

(ALSO HELPS WITH THE ACTUAL JOB)

Meet your multi-disciplinary team

Amanda Di Ienno:
Course Coordinator
Professional Teaching Fellow
ENGGEN since 2020
10+ years in Chemical Engineering and Systems
Keeps us in line!

Juliet Gerrard, Course Director
Professor in Science and Engineering
2018 – 2024, PM's Chief Science Advisor
DPhil (Biological Chem, Oxford)
BA (Hons) (Chemistry, Oxford)
Not an engineer!

Marc Lewis:
Course Coordinator
Professional Teaching Fellow,
ENGGEN since 2022
BE (Hons)/BCom,
PhD candidate CHEMMAT
Is an engineer!

Special mystery guest contributors

Peter Rachor
Hynds Entrepreneurial Teaching Fellow,
Center for Innovation and Entrepreneurship

We will look at the administrative side of the course tomorrow

ENGGEN 403

Course Overview

Course Director
Professor Juliet Gerrard

Course Coordinator
Amanda Di Ienno
Marc Lewis

Need to contact us? Use the Contact Form

Lectures

- *Tuesday 1-2 pm*
- *Thursday 2-3 pm*
- *Friday 1-2 pm*

Week-long Projects

- *Team Project*
- *Systems Project*

Group Assessment (75%)

- *Team Project Report*
- *Systems Project Video & Report*

Individual (25%)

- *Individual Goal Setting*
- *Lecture Participation*
- *Quiz 1*
- *Quiz 2*
- *Team Peer Review 1 & 2*
- *Personal Reflection*

Systems Thinking → Solution Analysis → Project Execution → Systems Week

Today is about how you can get the best out of the course

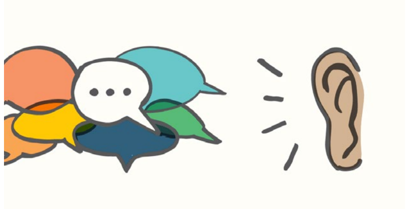
And to help you set your individual goals – your first assignment

<u>DUE:</u>

Monday JULY 27th at 10 pm

Outline

- The why - what students said last year
- Perspective from a recent graduate
- T-shape engineers
- Growth mindset
- Grit



What last year's students said …

positive: real world problems, clinics, cohort building, case studies
negative: the quizzes, political content

What we are doing about it …

positive: increased real world elements, committed to clinics, earlier in-person group interaction
negative: made the goals of the quizzes clearer, clearer format and decreased the weighting of quizzes, emphasis on role of politicians versus policy makers, less polarising examples

"Overall, the overarching lesson I've learned is that **professional growth often happens in the discomfort zone** and when we take small but deliberate actions to change ingrained habits. By linking my actions to measurable objectives, I became more mindful of how I engage with others and how my behaviour contributes to team outcomes. While not every goal was perfectly met, each one facilitated **deeper learning about communication, leadership, and collaboration - all of which are skills that extend beyond this course and into my broader professional development**."

"… the core insight was the direct application of systems thinking: the human elements (trust, communication, perceived professionalism) are inputs into the overall system's success. **I realised that my communication style was creating a negative feedback loop**; I needed to prioritize professional conduct and clear communication to establish my value and enable effective collaboration … "

“Lesson 1: **Coherence beats cleverness**

A solution only gains credibility when the strategic case, options analysis, CBA metrics, and rollout plan tell the same story; otherwise decision-makers fixate on inconsistencies.

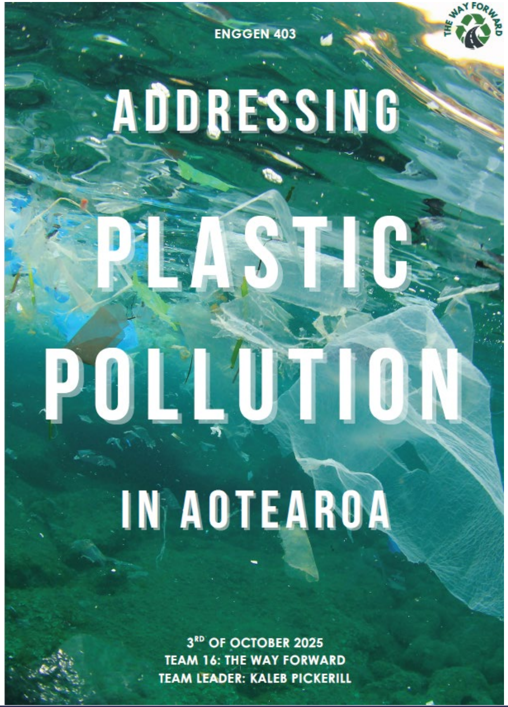
Lesson 2: **Evidence is a team sport**”

“I learned that **actually talking to people made everything easier** than Discord.

I made a friend and said hello to them at the gym, and it felt great.”

Recent graduate perspective

Kaleb Pickerill, BE(Hons)/BSc Engineering Science (2025), ENGGEN 403 Systems Week Team Leader



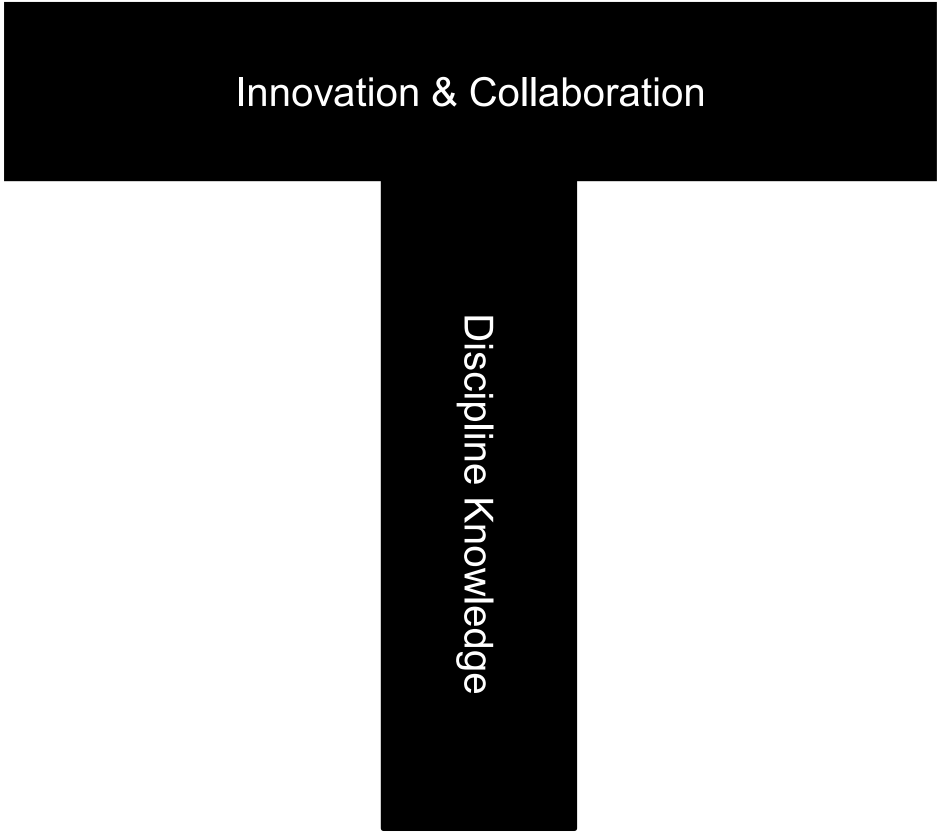
- come to lectures
- talk to people
- get to know your team as soon as you can ...

- students that do this get more out of the course
- are better prepared for the jobs they do next
- make a network that may last a career

Learning Outcomes

At the end of this session, students should be able to:

- Describe and apply the key concepts of a Growth Mindset
- Describe and apply the key concepts of Grit
- Consider how Growth Mindset and Grit can be leveraged and applied in your own life and aspirations
- Identify some areas in which you have a fixed mindset in which to adopt a Growth Mindset



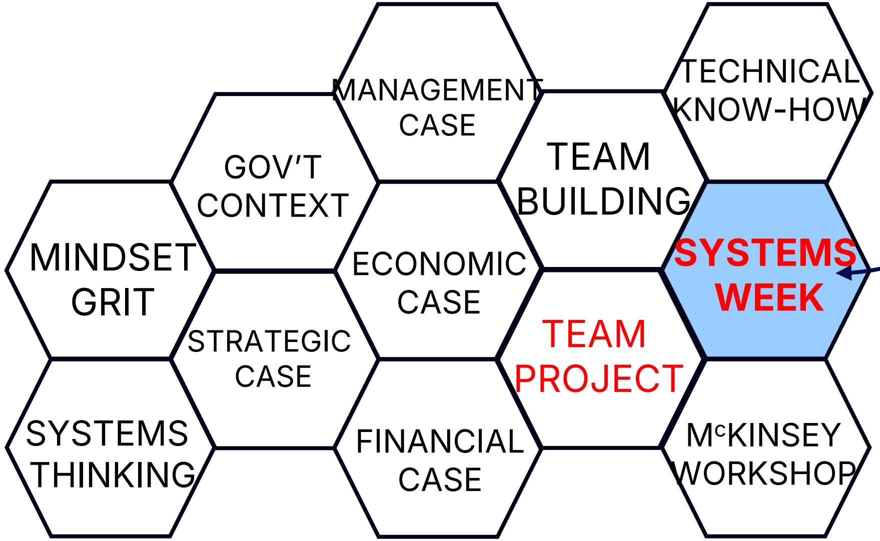
"T-shaped people " [have] a depth of knowledge in at least one discipline and a breadth of knowledge about innovation that allows them to work effectively with professionals of other disciplines to bring their ideas to life"

"Teaching engineering fundamentals in the classroom is important, but it's not enough," said Richard Miller of (US) National Academy of Engineering. "**Solving our planet's Grand Challenges requires engineering expertise, but they won't be solved by engineers alone**. Doubling down on even more hard sciences and math will not help. Instead, we need to incorporate new elements into engineering students' education to give them both **the skillset and the mindset needed to become leaders in addressing societal challenges.**"

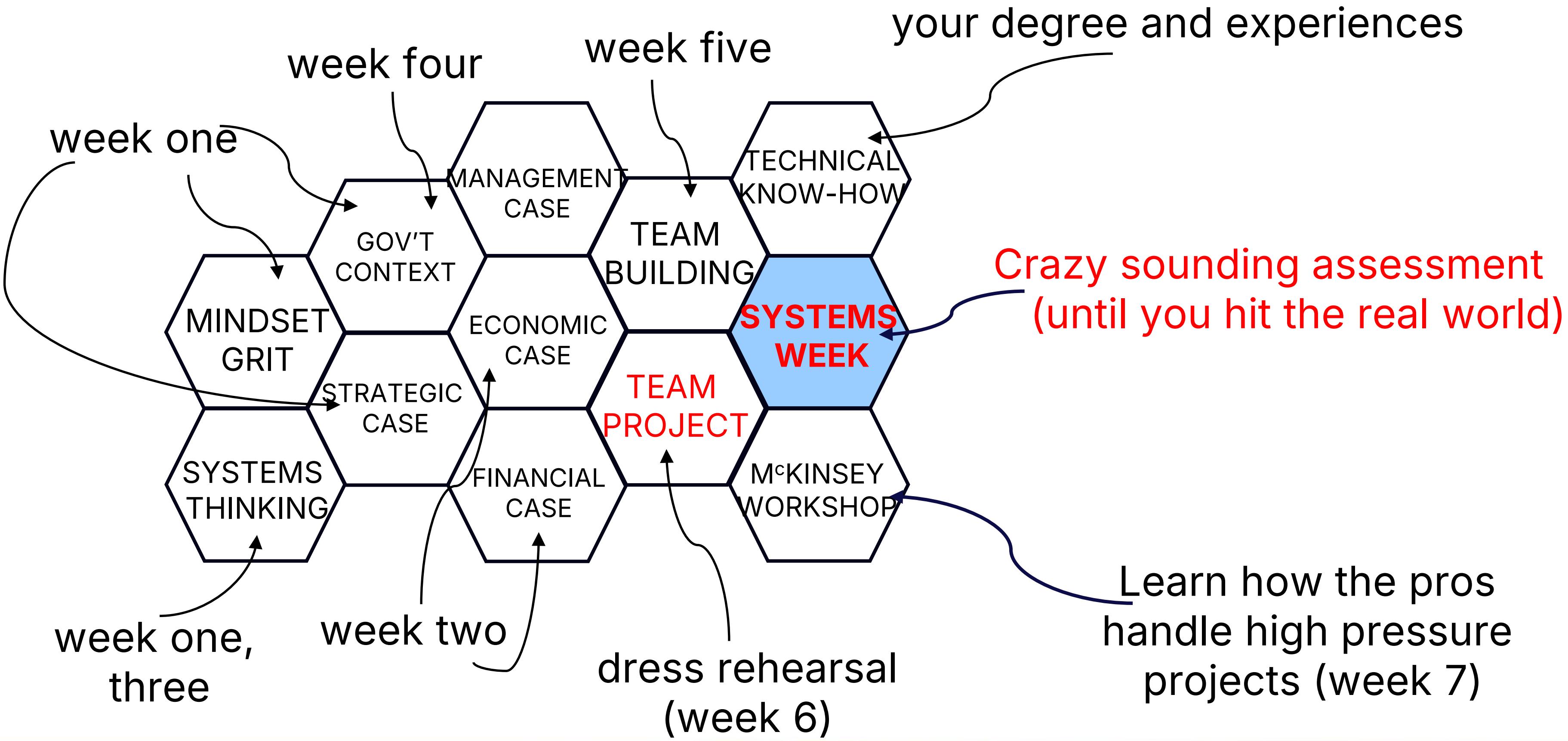
engineering
new zealand
te ao rangahau

Washington Accord

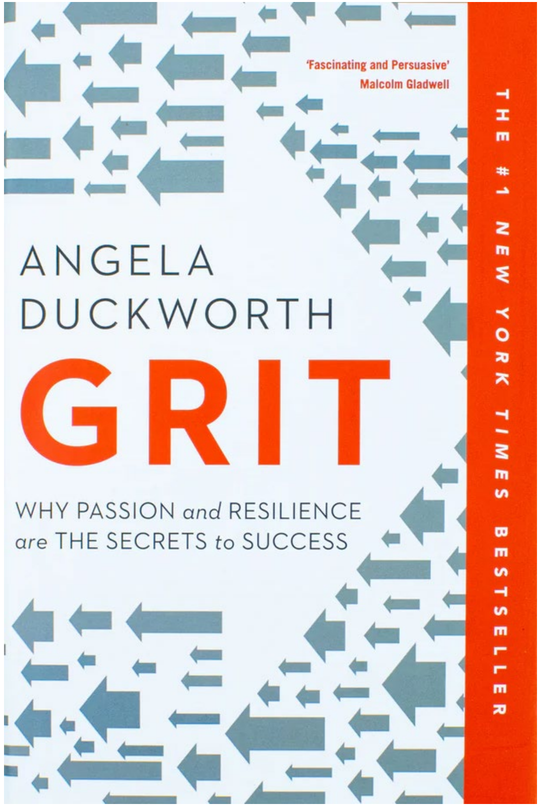
INTERNATIONAL
ENGINEERING
ALLIANCE



Crazy sounding assessment
(until you hit the real world)


your degree and experiences
week five
week four
week one
MANAGEMENT CASE
TECHNICAL KNOW-HOW
GOV'T CONTEXT
TEAM BUILDING
Crazy sounding assessment
(until you hit the real world)
MINDSET GRIT
ECONOMIC CASE
SYSTEMS WEEK
STRATEGIC CASE
TEAM PROJECT
SYSTEMS THINKING
FINANCIAL CASE
McKINSEY WORKSHOP
Learn how the pros handle high pressure projects (week 7)
week one, three
week two
dress rehearsal (week 6)


"Failure is an opportunity to grow"
GROWTH MINDSET
"I can learn to do anything I want"
"Challenges help me to grow"
"My effort and attitude determine my abilities"
"Feedback is constructive"
"I am inspired by the success of others"
"I like to try new things"
"Failure is the limit of my abilities"
FIXED MINDSET
"I'm either good at it or I'm not"
"My abilities are unchanging"
"I don't like to be challenged"
"I can either do it, or I can't"
"My potential is predetermined"
"When I'm frustrated, I give up"
"Feedback and criticism are personal
"I stick to what I know"
© Big Change



- To succeed you need passion and perseverance for your goals, over a sustained period of time
- Having talent doesn't mean you have grit – they are unrelated, maybe inversely related

- Four key pillars:

    1. Passion and interest
    2. Deliberate practice
    3. Sense of Purpose
    4. Sense of Hope

How to Grow Your Grit

"Learning to stick to something is a life skill that we all have to develop." – Angela Duckworth

Interest

"Whatever it is that you want to do, you'll find in life that if you're not passionate about what it is you're working on, you won't be able to stick with it." *– Jeff Bezos*

Angela used the question: "what makes people successful and why?" to stay interested in her work and get through graduate school.

What question will compel you to stay gritty?

"Every gritty person I've studied can point to aspects of their work they enjoy less than others, and most have to put up with at least one or two chores they don't enjoy at all. Nevertheless, they're captivated by the endeavor as a whole." – Angela Duckworth

Practice

"You must zero in on your weaknesses, and you must do so over and over again, for hours a day, week after month after year. To be gritty is to resist complacency. "Whatever it takes, I want to improve!" is a refrain of all paragons of grit, no matter their particular interest, and no matter how excellent they already are". *– Angela Duckworth*

Love the act of improvement, be better than you were yesterday and resist complacency.

The effort that you apply each day counts twice towards achievement. How? **Effort x Talent = Skill, Skill x Effort = Achievement**

Purpose

Angela surveyed over 16,000 Adult Americans and found the following to be true:

"Grittier people are dramatically more motivated than others to seek a meaningful, other-centered life. Higher scores on purpose correlate with higher scores on the Grit Scale." *– Angela Duckworth*

The desire to aid to the wellbeing of others is likely to sustain your interest and make you grittier.

How can you connect what you do to the well-being of others?

Hope

"Grit depends on a different kind of hope. It rests on the expectation that our own efforts can improve our future." *– Angela Duckworth*

Grit requires an enduring belief that your skills are malleable and not set in stone.

Modern science shows that our brains continue to grow well past childhood. Daily experience continuously shapes the adult brain.

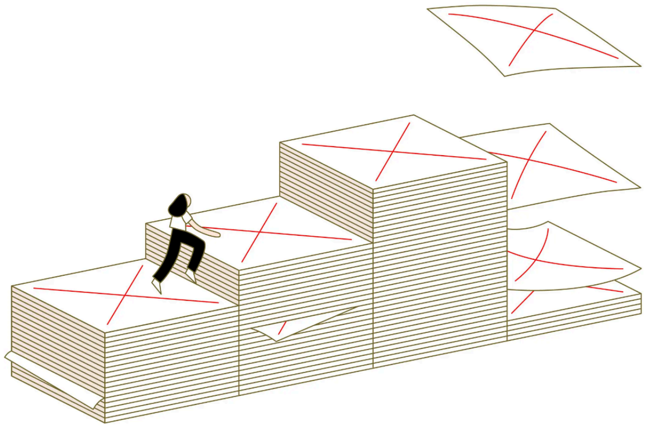
Ditch the idea of having fixed abilities and recall time when you learned something you thought you never could: driving a car, learning a new sport, learning a new software program, etc.

The New York Times

Do You Keep a Failure Résumé? Here's Why You Should Start.

Failure isn't a roadblock. It's part of the process.

Share full article



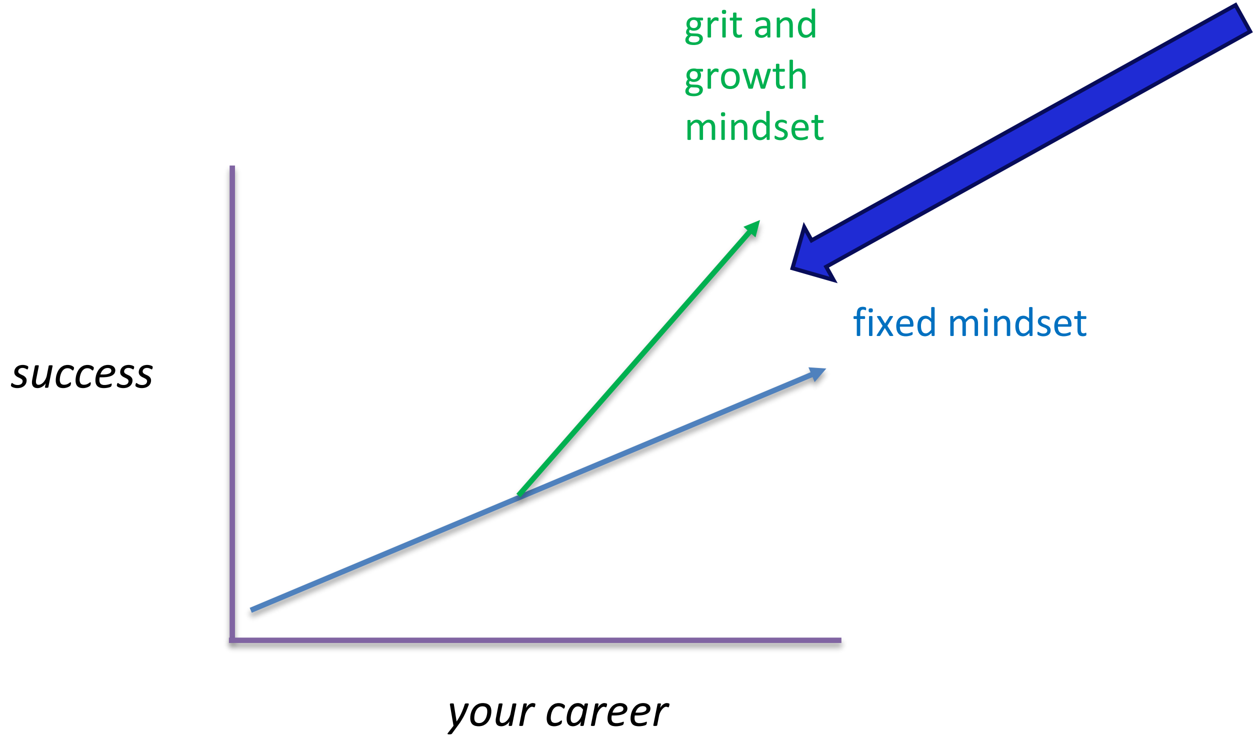
Rose Wong

- Perseverance, resilience, mental toughness and determination leads to success in achieving life goals

- Ability to delay instant gratification in favour of long-term success

- People with high levels of grit are better able to find solutions when problems and challenges arise

- You will need to bring all your grit in system😉ek



ENGGEN 403 provides this development opportunity – set goals to enable you to get better at something you might need in your workplace

Questions?

Remember – your first assignment on individual goals is due on MONDAY

(JULY 27th at 10 pm)